



DIPARTIMENTO DI ELETTRONICA,
INFORMAZIONE E BIOINGEGNERIA

Politecnico di Milano

Machine Learning (Code: 097683)

February 15, 2022

Surname:

Name:

Student ID:

Row:

Column:

Time: 2 hours

Maximum Marks: 33

- The following exam is composed of **8 exercises** (one per page). The first page needs to be filled with your **name, surname and student ID**. The following pages should be used **only in the large squares** present on each page. Any solution provided either outside these spaces or **without a motivation** will not be considered for the final mark.
- During this exam you are **not allowed to use electronic devices**, such as laptops, smartphones, tablets and/or similar. As well, you are not allowed to bring with you any kind of note, book, written scheme, and/or similar. You are also not allowed to communicate with other students during the exam.
- The first reported violation of the above-mentioned rules will be annotated on the exam and will be considered for the final mark decision. The second reported violation of the above-mentioned rules will imply the immediate expulsion of the student from the exam room and the **annulment of the exam**.
- You are allowed to write the exam either with a pen (black or blue) or a pencil. It is your responsibility to provide a readable solution. We will not be held accountable for accidental partial or total cancellation of the exam.
- The exam can be written either in **English** or **Italian**.
- You are allowed to withdraw from the exam at any time without any penalty. You are allowed to leave the room not early than half the time of the duration of the exam. You are not allowed to keep the text of the exam with you while leaving the room.
- **Three of the points will be given on the basis on how quick you are in solving the exam.** If you finish earlier than 45 min before the end of the exam you will get 3 points, if you finish earlier than 30 min you will get 2 points and if you finish earlier than 15 min you will get 1 point (the points cannot be accumulated).
- The box on Page 10 can only be used to complete the Exercises 7, and/or 8.

Ex. 1	Ex. 2	Ex. 3	Ex. 4	Ex. 5	Ex. 6	Ex. 7	Ex. 8	Time	Tot.
/ 7	/ 7	/ 2	/ 2	/ 2	/ 2	/ 4	/ 4	/ 3	/ 33

Exercise 1 (7 marks)

Explain what is the **VC dimension** of a hypothesis space and what it is used for.

Exercise 2 (7 marks)

Describe the **PCA** technique and what it is used for.

Exercise 3 (2 marks)

Consider the following snippet of code

```
1 x = zscore(dataset['petal-length'].values).reshape(-1, 1)
2 y = zscore(dataset['petal-width'].values)
3
4 Phi = np.ones((len(x), 2))
5 Phi[:, 1] = x.flatten()
6 model = inv(Phi.T @ Phi) @ (Phi.T.dot(y))
7 r_model = linear_model.Ridge(alpha=10)
8 r_model.fit(x, y)
9 l_model = linear_model.Lasso(alpha=10)
10 l_model.fit(x, y)
```

1. Which problem is the snippet above solving?
2. Which algorithms are implemented by the snippet above?
3. Which model would you choose to provide an interpretable solution, e.g., a model whose parameters influence are easy to explain?
4. Which are the pros and the cons of using the algorithm in lines 4-6?

Exercise 4 (2 marks)

We want to monitor the presence of a contaminant in the water distribution systems of a large city. We are provided with the choice of a fixed number L of different locations where to monitor the contaminant concentration c_l , but we are constrained to select a single location per day, due to shortage of personnel. Our goal is to maximize the average concentration of contaminant measured over a finite period T .

1. Model the previous problem using a Machine Learning approach (tell which kind of problem it is and specify the input/output or state/action/reward). Propose a method to solve the above problem.
2. The authorities are not convinced with your choice. Provide some theoretical evidence that the chosen method is sound and effective.
3. What would be the performance of an approach that selects the location each day randomly?

1. The problem represents a MAB problem with L different arms over a time horizon of T days. The reward consists in the contaminant concentration c_l collected, therefore minimizing the regret is equivalent to maximize the average concentration collected. If we assume that the environment does not change over time, we might resort to stochastic MAB techniques, such as UCB1 or TS.
2. The methods provides theoretical guarantees on the upper bound on the regret. Specifically, both the mentioned algorithms provide a regret of $O(\frac{\log(T)}{\Delta})$, where Δ is the average gap in terms of concentration w.r.t the optimal arms.
3. In this case each arm has the same chance of being pulled, therefore we would have a regret of $O(T\Delta)$.

Exercise 5 (2 marks)

Tell if the following properties holds for the Logistic Regression (LR) and Perceptron (PE) methods. Motivate your answer.

1. The training process is guaranteed to terminate in a finite amount of time.
2. The choice of the learning rate is crucial to speed up the learning process.
3. It is possible to determine the significance of the parameters learned by the method.
4. If the method converges, it may converge to different optimal parameter vectors.

1. True for LR since the loss function is convex, false for PE if the classes are not linearly separable.
2. True for LR since we are applying an online gradient descend optimization approach to minimize the loss, false for PE since it gradually decrease its influence over time.
3. True for LR where the parameters have significance for the log odds of the model, false for PE which only provides a sharp $\{0, 1\}$ label.
4. False for LR since the loss function is convex and provides a single optimal parameter vector, true for PE since, if the classes are linearly separable, there may be multiple boundaries separating the training data perfectly.

Exercise 6 (2 marks)

Tell if the following statements about parametric and non-parametric methods are true or false. Motivate your answers.

1. To address a classification task on a large dataset of low-dimensional points, it is usually better to employ a non-parametric method than a parametric one.
2. When a regression task requires to provide real-time predictions, it is in general a good idea to train a non-parametric method.
3. Non-parametric methods are generally less affected by the curse of dimensionality than parametric methods.
4. The Bayesian linear regression is a non-parametric method.

1. FALSE, non-parametric methods are usually convenient when data are high-dimensional, whereas parametric methods can better handle very large low-dimensional datasets.
2. FALSE, non-parametric methods (e.g., k-nearest neighbors) are usually slower in prediction than parametric methods (e.g., linear regression).
3. TRUE, non-parametric methods can generally deal with an exponentially-growing feature space through the kernel trick.
4. FALSE, Bayesian linear regression is a parametric method, in which the parameters are learned from data through a Bayesian approach.

Exercise 7 (4 marks)

Consider a linear, hard-margin, two-class SVM classifier defined by parameters:

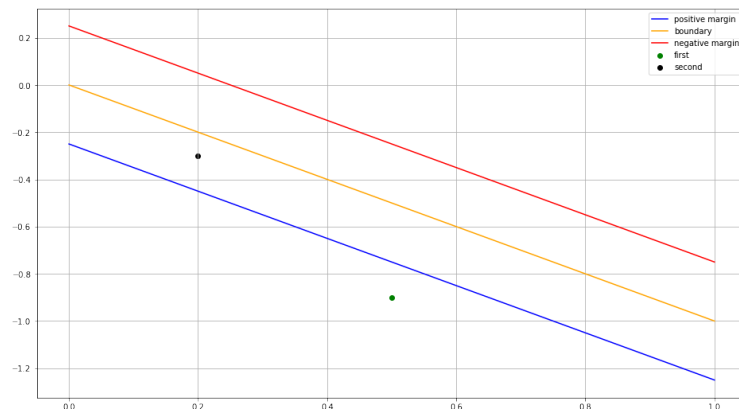
$$w = [-4, -4], b = 0.$$

Answer the following questions providing adequate motivations.

1. Provide the analytical formula of the boundary and the margins.
2. How is the point $p_1 = [0.5; -0.9]$ classified according to the trained SVM?
3. Assume to collect a new sample $p_2 = [0.2; -0.3]$ in the negative class. Do you need to retrain the SVM?

1.

$$\begin{aligned} -4x_1 + -4x_2 + 0 &= 0 && \text{(boundary)} \\ -4x_1 + -4x_2 + -1 &= 0 && \text{(positive margin)} \\ -4x_1 + -4x_2 + 1 &= 0 && \text{(negative margin)} \end{aligned}$$



2.

$$-4 \cdot 0.5 + -4 \cdot -0.9 + 0 = 1.6 \rightarrow \text{positive}$$

3.

$$-4 \cdot 0.2 + -4 \cdot -0.3 + 0 = 0.4 \rightarrow \text{positive}$$

It would be misclassified, thus you need to retrain and p_2 is a SV. You would use a soft-margin if the problem is no more linearly separable.

Exercise 8 (4 marks)

Consider the following episode obtained by an agent interacting with an MDP having two states $\mathcal{S} = \{A, B\}$ and two actions $\mathcal{A} = \{l, r\}$,

$$(A, l, 1) \rightarrow (A, l, 1) \rightarrow (A, r, 0) \rightarrow (B, r, 10) \rightarrow (B, l, 0) \rightarrow (A, r, 0) \rightarrow (B, l, 0) \rightarrow (A).$$

Answer to the following questions providing adequate motivations.

1. Execute the *Q-learning* algorithm on the given episode considering initial state-action values $Q(S, a) = 0$ for every state-action pair, learning rate $\alpha = 0.5$, and discount factor $\gamma = 0$.
2. Provide the best policy according to the output of *Q-learning*.
3. Do you think that the agent fully exploited the policy learned in the episode above? Make a consistent guess with the available information.

1. We start with the initial values $Q(A, l) = Q(A, r) = Q(B, l) = Q(B, r) = 0$. Then, we compute the *Q-learning* update $Q(S_t, a_t) = (1 - \alpha)Q(S_t, a_t) + \alpha(R_t + \gamma \max_{a \in \{l, r\}} Q(S_{t+1}, a))$ for every step t in the episode:

- (a) $Q(A, l) \leftarrow 0.5Q(A, l) + 0.5(1 + 0) = 0.5 \cdot 0 + 0.5 \cdot 1 = 0.5$
- (b) $Q(A, l) \leftarrow 0.5Q(A, l) + 0.5(1 + 0) = 0.5 \cdot 0.5 + 0.5 \cdot 1 = 0.75$
- (c) $Q(A, r) \leftarrow 0.5Q(A, r) + 0.5(0 + 0) = 0.5 \cdot 0 + 0.5 \cdot 0 = 0$
- (d) $Q(B, r) \leftarrow 0.5Q(B, r) + 0.5(10 + 0) = 0.5 \cdot 0 + 0.5 \cdot 10 = 5$
- (e) $Q(B, l) \leftarrow 0.5Q(B, l) + 0.5(0 + 0) = 0.5 \cdot 0 + 0.5 \cdot 0 = 0$
- (f) $Q(A, r) \leftarrow 0.5Q(A, r) + 0.5(0 + 0) = 0.5 \cdot 0 + 0.5 \cdot 0 = 0$
- (g) $Q(B, l) \leftarrow 0.5Q(B, l) + 0.5(0 + 0) = 0.5 \cdot 0.5 + 0.5 \cdot 0 = 0.25$

which results in $Q(A, l) = 0.75, Q(A, r) = 0, Q(B, l) = 0, Q(B, r) = 5$.

2. We select the greedy policy $\pi(S) \in \arg \max_{a \in \{l, r\}} Q(S, a)$ w.r.t. the state-action values obtained with *Q-learning*, which gives $\pi(A) = l, \pi(B) = r$.
3. Since at the steps (c), (e), (f), (g) the agent is not choosing the action that is maximizing the current estimate of the *Q* values, we can infer that the episode does involve some exploration.

